

Still Learning

Complete Product Documentation

Version 2.1 | July 2026

Stack: Python 3.11+ / FastAPI / SQLAlchemy / bcrypt + JWT

URL: <https://still-learning.onrender.com>

1. Complete System Architecture

Stack: Python 3.11+ / FastAPI / SQLAlchemy / SQLite (dev) / PostgreSQL (prod) / bcrypt + JWT. The system follows a monolithic architecture with lightweight server-side rendering and JavaScript on the client for SPA-like navigation.

Layers

- Client: HTML + CSS + Vanilla JS, PWA with Service Worker and Manifest
- Server: FastAPI with Uvicorn, 55 REST endpoints
- Data: SQLAlchemy ORM, 16 tables, automatic migrations
- Auth: bcrypt for passwords, JWT (HS256, 48h) for sessions
- Deploy: Render (free tier), PostgreSQL in production

Directory Structure

```
project quiz code/  
main.py 55 endpoints (1893 lines)  
models.py 16 ORM models  
database.py SQLAlchemy Engine + Session  
translations.py i18n PT/EN (431 lines)  
static/ CSS, PWA manifest, sw.js, SVG icons  
templates/ 6 HTML templates  
docs/ PDF documentation
```

2. Navigation Flow

User logs in at /. If 'couple' profile, redirects to /still-learning with 3-item bottom nav: Today (smart dashboard), Us (sub-menus: diary, memories, quiz, challenge), and Gear (settings/language/logout). If 'quiz' profile, starts a 10-question session: /start → /quiz → /result.

SPA Internal Nav

- Today: Dashboard, Question of the Day, Challenge, Quote, Agenda, To-Dos, Weekly Review
- Us: Diary, Memories, Couple Quiz, Challenge
- Gear: Language (PT/EN), Logout (clear localStorage)

Authentication

Login → JWT stored in localStorage → api() sends Authorization: Bearer <token> on every request → require_auth() validates JWT (HS256, 48h) on the backend.

3. User Journey

First Access

- User accesses still-learning.onrender.com and logs in
- System verifies bcrypt (with legacy SHA-256 fallback)
- If 'couple': generates 48h JWT and redirects to /still-learning with empty dashboard
- If 'quiz': starts new 10-question session or resumes incomplete session

Daily Journey

- Morning: Opens Today → sees greeting, stats, daily quote
- Afternoon: Opens Question of the Day → answers and waits for partner
- Evening: Opens Us → Diary → writes about the day with mood

Weekly Journey

- Weekly Review with reflection
- Varied challenges: riddle, intimate question, photo, quote, custom challenge

4. Text Wireframes

The interface follows a mobile-first layout with max-width 520px, fixed header with title and language button, central content in cards with pink shadow, and fixed bottom nav with 3 items. Colors: pastel pink (#ffb5c2), off-white (#fdf6f0) and dark gray text (#4a4a4a).

Main Screens

- Login: Rounded inputs, SVG heart, gradient button, PT/EN toggle
- Today: Greeting card, stats (pills), question, diary, challenge, quote, agenda
- Us: 2x2 grid of Diary, Memories, Quiz, Challenge cards
- Settings (Modal): Semi-transparent overlay with Language and Logout options
- Question of the Day: Question card, textarea, send button, partner's answer
- Diary: 'Today' section with own entry + timeline of previous days

5. Page Structure

Base Template

```
<!DOCTYPE html>
<html lang="pt-BR">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width">
<meta name="theme-color" content="#ffb5c2">
<link rel="manifest" href="/static/manifest.json">
<link rel="stylesheet" href="/static/style.css">
</head>
<body>
<div class="floating-bg"></div>
</body>
</html>
```

Templates

- index.html: Login (entry point)
- quiz.html: Interactive quiz (10 questions)
- result.html: Quiz result with personalized message
- retry.html: Retry after completion
- couple.html: Main SPA-like app (~2250 lines)
- admin.html: Admin panel (~1800 lines)

6. Reusable Components

Frontend (Vanilla JS)

- `api(path, opts)`: Fetch wrapper with JWT + headers + auto JSON
- `showSection(name)`: SPA navigation (swaps content + active nav)
- `backLink()`: Standardized 'Back' button
- `esc(s)`: HTML sanitization (textContent + attribute escaping)
- `attrEsc(s)`: HTML attribute sanitization (quote escaping)
- `toggleLang()`: PT/EN switch via cookie + reload
- `logout(e)`: Clear localStorage + redirect to /
- `toast(msg)`: Non-intrusive floating toast (3s)
- `markDirty()`: Marks admin form as unsaved
- `initDecorations()`: Animated background with floating emojis and stars

Backend (Python)

- `_render(name, **kwargs)`: Custom template engine (string replace)
- `require_auth()`: JWT auth dependency injection (Bearer only)
- `check_admin()`: Admin credential validation (ADMIN_USERS env var)
- `get_partner_name()`: Resolves partner name in a couple
- `hash_pw()` / `verify_pw()`: bcrypt hash/verify + SHA-256 fallback
- `create_jwt()` / `decode_jwt()`: JWT HS256 generation/validation

7. Design System

CSS Tokens

```
:root {
  --pink: #ffb5c2; /* Primary */
  --pink-dark: #e88a9a; /* Hover/Active */
  --cream: #fff5f5; /* Surface */
  --text: #4a4a4a; /* Main text */
  --text-light: #b0a0a0; /* Secondary text */
  --bg: #fdf6f0; /* Background */
  --radius-sm: 10px;
  --radius-cloud: 18px;
  --shadow: 0 4px 20px rgba(255,181,194,0.2);
  --ease-bounce: cubic-bezier(0.34,1.56,0.64,1);
}
```

Typography

- Titles: 'Press Start 2P', monospace (9px)
- Body: -apple-system, BlinkMacSystemFont, sans-serif (0.9rem)

Icons

Exclusive use of Unicode emoji: 📅 Today, 👤 Us, ⚙️ Settings, ❤️ Love, ❓ Question, 📖 Diary, 📁 Memories, 🏆 Challenge, 🎮 Quiz, 📅 Agenda, 📝 To-Dos, 💬 Quote, 🔔 Notification.

8. Visual Guide

Pastel pink palette with off-white. Mobile-first layout (max-width: 520px). Cards with soft shadow, rounded corners (10-18px). Fixed header with title and language button. Fixed bottom nav with 3 items. Transitions with cubic-bezier bounce for organic feel.

Responsiveness

- Desktop (>768px): Centered at 520px
- Mobile (<768px): Full width with 16px padding
- Small mobile (<375px): Font-size reduced 10-15%

9. Database Structure

16 tables in SQLAlchemy ORM. SQLite in dev, PostgreSQL in prod.

Tables

- profiles: User profiles (quiz or couple)
- login_credentials: Login + password_hash (bcrypt)
- quiz_sessions: Quiz sessions with JSON answers
- login_events: Login audit trail
- couples: User pairs (user1_id, user2_id)
- diary_entries: Diary entries by date
- daily_questions: Daily questions with answers (answer_a, answer_b)
- challenges: Challenges with 5 rotating types
- agenda_events: Agenda events with date/time
- todo_items: Todo items with done status
- weekly_reviews: Weekly reviews (reflection_a, reflection_b)
- quiz_answers: Couple quiz answers (about_self, about_partner)
- quote_refreshes: Quote navigation state
- photos: Base64 photos with caption

10. APIs

Total: 55 documented REST endpoints. Authentication via JWT Bearer exclusively.

Auth (1 endpoint)

- POST /api/login: Login with name+password, returns JWT token + user data

Quiz (5 endpoints)

- POST /api/start: Start quiz session
- GET /api/question/{id}: Current question
- POST /api/answer/{id}: Submit answer
- GET /api/result/{id}: Complete result with score and message
- POST /api/result/final/{id}: Final romantic response

Couple App (28 endpoints)

- Dashboard, Question of the Day, Diary (CRUD)
- Challenges: riddle, intimate question, photo, quote, custom (+ delete)
- Agenda (add + delete), To-Dos (add + toggle + delete), Weekly Review
- Quotes, Couple Quiz, Memories

Admin (21 endpoints)

- CRUD for couples, profiles, questions, challenges, photos, diaries, agenda, todos
- Statistics, login history, quiz results
- Couple data reset/delete (with parallel batch delete)

11. AI Structure

Current state: No generative AI. All content is predefined: 40 daily questions, 41 themes, 18 challenge suggestions, 30 quotes with trivia. Rotation is deterministic based on days since couple creation. The quiz uses `random.shuffle()` to order 10 of 40+ available questions.

Future Opportunities

- Personalized questions via LLM (GPT-4 / Claude)
- Sentiment analysis for diary entries (NLP)
- Automatic weekly summary via LLM
- Quote recommendation by mood (embeddings + similarity)
- Adaptive quiz with questions the partner got wrong
- Communication pattern detection

12. Development Roadmap

v1.0 - MVP (Current)

- Login bcrypt + JWT, Dashboard, Question of the Day, Diary
- Challenges (5 types: riddle, photo, intimate question, custom, quote)
- Agenda, To-Dos, Weekly Review, Couple Quiz, Quotes, Memories
- Admin (CRUD couples/profiles, statistics)
- PWA (Service Worker + Manifest), i18n PT/EN
- 3-item bottom nav (Today/Us/Gear)

v2.0 - Connectivity

- Push notifications (Web Push API), Streak tracking
- Dark mode, Automated tests (pytest + Playwright)
- CI/CD (GitHub Actions)

v3.0 - AI & Personalization

- Smart suggestions, Diary sentiment analysis
- Automatic weekly summary, Adaptive quiz, Contextual chat

v4.0 - Social & Expansion

- Single/friends mode, Shared relationship goals
- Yearly digital book, PDF export, Public API

13. Launch Roadmap

- Pre-launch: Test full local flow (1 week)
- Alpha: 2-3 couple friends use for 2 weeks (D7 retention > 80%)
- Closed beta: 10-20 couples with structured feedback (NPS > 40)
- Open beta: WhatsApp/Telegram group promotion (100 couples)
- v1.0 Launch: Instagram, Product Hunt (500 active users)
- v2.0: Push + Streaks (2 months post-launch, DAU > 20% MAU)
- v3.0: AI (4 months post-launch, avg time > 8min/day)

14. Growth Strategy

Acquisition Channels

- Word of mouth (couples refer couples) - Free, very high potential
- Instagram - daily question posts with link in bio
- TikTok - couple challenge videos
- WhatsApp/Telegram couple groups
- Partnerships with couple therapists and relationship influencers

Viral Cycle

Couple A uses → Takes screenshot of a question/dashboard → Posts on Instagram/TikTok → Couple B sees it → Wants to try → Couple B invites partner → Cycle repeats.

15. Monetization Strategy

Freemium Model

- Free: Basic features (1 challenge/day, 3 photos/day, last 30 memories)
- Monthly Premium (R\$9.90): Unlimited challenges, unlimited photos, all memories, sentiment analysis, dark mode
- Annual Premium (R\$79.90/year): 33% discount
- Lifetime (R\$199.90): Early adopters, launch price

Payment Methods

- Stripe (international cards)
- Mercado Pago / Pix (Brazil)
- Apple Pay / Google Pay

16. Retention Strategy

Daily Triggers (Push)

- 08:00 - 'Question of the Day available!'
- 12:00 - 'Your partner answered the question!'
- 18:00 - 'Time for today's challenge!'
- 21:00 - 'How about writing in the diary?'

Gamification

- Streaks: Consecutive days of use (fire)
- Achievements: '7 days of questions', 'First photo together'
- Couple Levels: Bronze → Silver → Gold → Diamond
- Monthly Badges: 'February Full of Love', 'March of Connection'

Churn Prevention

- 3 days without login: Email 'We miss you!'
- 7 days: Notification with special question waiting
- 14 days: 'Want a break? Your data is safe'

17. Improvements Checklist

P0 - Critical

- ☒ [OK] Migrate from on_event() to lifespan events
- ☐ [] Add automated tests (pytest for API, Playwright for frontend)
- ☐ [] Rate limiting (brute-force protection)
- ☐ [] Photo upload validation (size, MIME type)
- ☐ [] Pagination on all lists (memories, admin)

P1 - High

- ☒ [OK] Admin XSS fixed (11 locations)
- ☒ [OK] esc() improved (single quote escaping)
- ☒ [OK] Dirty tracking in admin (beforeunload + markDirty)
- ☒ [OK] Parallel batch delete with progress feedback
- ☐ [] Dark mode with persistent toggle
- ☐ [] Streak tracking (consecutive days)
- ☐ [] Real push notifications (Web Push API)
- ☐ [] Configurable relationship start date

P2 - Medium

- ☐ [] 'Surprise me' button on dashboard
- ☐ [] Offline mode (last session cache)
- ☐ [] Keyboard shortcuts (1-4 for navigation)
- ☐ [] Export history as PDF

18. Prioritized Backlog

Sprint 1 - Foundation (Now)

- ☒ [OK] Migrate lifespan events, Admin UX, XSS fixes
- ☐ [] Automated tests (2 days, P0)
- ☐ [] Rate limiting (4 hours, P0)
- ☐ [] Upload validation (2 hours, P1)

Sprint 2 - Retention

- ☐ [] Streak tracking (1 day, P1)
- ☐ [] Day counter (3 hours, P1)
- ☐ [] Push notifications (3 days, P1)

Sprint 3 - Engagement

- ☐ [] Dark mode (1 day, P1)
- ☐ [] Pagination (1 day, P2)

Sprint 4 - Monetization

- ☐ [] Free/premium plan system (5 days, P1)

- [] Stripe + Mercado Pago (6 days, P1)

19. Product Evolution Plan

Phase 1 - Comfort (Months 1-3)

Focus on stability and retention: finish technical debt, implement streaks + day counter, improve mobile experience (gestures, haptics), collect alpha/beta user feedback.

Phase 2 - Connection (Months 3-6)

Focus on product depth: push notifications, dark mode, more challenge types (video, audio, location), more quiz categories, basic sentiment analysis.

Phase 3 - Personalization (Months 6-9)

Focus on AI and individualization: LLM integration for personalized questions, automatic weekly summary, history-based recommendations, activity suggestions.

Phase 4 - Expansion (Months 9-12)

Focus on growth and monetization: premium plan, friends and individual mode, public API, couple digital book, native app (React Native).

20. Risk List

- Data loss (Low/Catastrophic): PostgreSQL backup + manual export
- Intimate data leak (Low/Catastrophic): HTTPS, encryption at rest
- Login brute-force (Medium/High): Rate limiting + slow bcrypt by design
- Churn after 7 days (High/High): Better onboarding, notifications, gamification
- Low adoption (High/High): Growth hacking + therapist partnerships
- Competition (Lovewick, Couply) (Medium/Medium): Differentiators: diary + affordable pricing
- LLM cost (AI) (Medium/Medium): Cache, static content fallback, batch processing

21. Metrics (KPIs)

Product

- MAU: 500 (3 months) / 2,000 (6 months)
- DAU: 100 (3 months) / 500 (6 months)
- DAU/MAU: 20% (3 months) / 25% (6 months)
- D1 Retention: 60% | D7: 30% | D30: 15%
- Avg session time: 4 min (3m) / 6 min (6m)

Engagement (per couple/week)

- Questions answered: 4 of 7
- Challenges completed: 3 of 5
- Diary entries: 3
- Average streak: 7 consecutive days

Business

- Free → premium conversion rate: 5%
- LTV (Lifetime Value): R\$60
- MRR (Monthly Recurring Revenue): R\$5,000 (1 year)
- NPS (Net Promoter Score): 50+
- Monthly churn rate: < 5%

Technical

- Uptime: 99.9%
- P95 response time: < 500ms
- Lighthouse Performance: > 85
- Lighthouse PWA: > 90

22. A/B Testing Plan

Test 1: Onboarding Flow

- A (control): Direct login → dashboard
- B (test): Login → guided tour (3 slides) → dashboard
- Hypothesis: guided tour increases D1 retention by 15%

Test 2: Notifications

- A: No notifications | B: 1 notif/day | C: 2 notifs/day
- Hypothesis: 2 notifications increase DAU by 30%

Test 3: Premium Pricing

- A: R\$9.90/mo, R\$79.90/yr | B: R\$14.90/mo | C: R\$9.90/mo, R\$69.90/yr

Test 4: Dashboard Layout

- A: Vertical cards (current) | B: 2x2 grid with larger summaries

23. Accessibility Plan

Target: WCAG 2.1 Level AA.

- Add aria-label to Gear button, lang toggle, and clickable cards
- Add role='navigation' to bottom nav and aria-current='page'
- Add role='dialog' to settings modal
- Ensure minimum contrast 4.5:1 for normal text
- Support prefers-reduced-motion for animations
- Tests: Lighthouse Accessibility, axe DevTools, VoiceOver, NVDA

24. Security Plan

Implemented

- bcrypt passwords (salt + 12 round cost)
- JWT with 48h expiration, HS256 signing (no legacy header fallback)
- SHA-256 fallback with automatic bcrypt upgrade on login
- Output sanitization with `esc()` + `attrEsc()` (XSS prevention in text and attributes)
- Single quote escaping in `esc()`
- Admin XSS fixed (11 locations)
- Dirty tracking (`beforeunload` + `markDirty`)
- Parallel batch delete with progress feedback
- SQLAlchemy ORM prevents SQL injection
- Mandatory HTTPS (Render provides TLS)
- CORS middleware configured
- Connection pool (`pool_size=10`, `max_overflow=20`)
- Service Worker caches only GET requests
- ADMIN_USERS via env var (no hardcoded fallback)

Pending

- Rate limiting (`slowapi` or custom middleware)
- Content Security Policy (strict CSP)
- Upload validation (5MB max, MIME type)
- Helmet-like headers (`X-Frame-Options`, `X-Content-Type-Options`)
- Dependency auditing (`pip-audit`, `Dependabot`)

25. SEO Plan

On-Page

- Meta description: 'Still Learning: the app for couples to connect every day'
- Open Graph tags (`og:title`, `og:description`, `og:image`)
- Structured Data (Schema.org `WebApplication`)
- Sitemap.xml and robots.txt
- Descriptive alt text on all SVG images

Target Keywords

- app for couples, couple app, questions for couples
- shared couple diary, challenges for couples
- strengthen relationship, couple quiz

26. Internationalization Plan

Current Status

- Portuguese (BR): Complete
- English (US): Complete
- Spanish: Pending | French: Pending | Italian: Pending | German: Pending

Translation Architecture

Nested dictionary: `L['key']['pt']` and `L['key']['en']`. `t(key, lang)` function extracts the correct value. Interface uses `getLang()` from cookie 'lang'.

Localization Roadmap

- Spanish: Maximum priority, 2 days effort
- French: High priority, 2 days
- Italian: Medium priority, 2 days
- German: Low priority, 2 days

Auto-Detection

Current: 'lang' cookie set manually by user. Proposed: Auto-detection via Accept-Language header + manual fallback.